PAYVAULT

SMART SME FINANCING AND GROWTH PLATFORM

No signal.
Still paid.

Offline payments that turn a market trader's daily sales into the credit history a bank can lend against.

Live demo: payvault.rumptycloud.app



THE PROBLEM

Africa's traders are creditworthy but invisible.

- **Payments stop when the signal stops.** In markets with weak networks, transfers fail, so sales fall back to cash.
- **Cash leaves no record.** Banks can't see a trader's real income, however steady it is.
- **No record, no credit.** Women-led micro-businesses are refused loans or pushed to costly informal lenders.
- **Growth stalls.** On a slow day she can't restock, and a missed day is lost income.



TARGET USERS

Built for the trader. Sold to the institutions that serve her.

1

Market traders

Women-led micro and small businesses: market stalls, kiosks, food sellers, agents. Smartphone or basic phone.

2

Their customers

Shoppers who want to pay digitally even where data is patchy, including on basic phones via USSD.

3

Partners (who pay)

Banks, microfinance banks, mobile money operators and fintechs. They hold the licences and funds; PayVault is the layer underneath.

Starting market: Nigeria. Designed for 15 African markets in English and French.

THE SOLUTION

PayVault: payments that work offline, and sales that become credit.

Pays offline

The customer's phone signs a payment with no network. The trader's phone verifies it on the spot. It settles automatically when any phone reconnects.

Sales become credit

Every settled sale counts as a trading day. A steady month of trading lifts her score and unlocks an overdraft from the partner bank.

Works on any phone

Web app for smartphones; USSD menu, SMS confirmations and a printed stall sticker (QR / NFC) for everyone else.

HOW A SALE WORKS

From a stall with no signal to a loan offer.

1

Load the vault

Once online, the customer moves money into an offline vault on her phone, plus an optional overdraft.

2

Pay with no signal

She scans the trader's code and enters her PIN. The phone signs the payment; the trader's phone verifies it in under a second.

3

Settle automatically

When either phone, or any nearby PayVault phone, gets signal, the payment settles and the trader gets an SMS.

4

Unlock credit

Settled sales build her trading history. The partner bank sees verified income and offers an overdraft.

INNOVATION

What's new about PayVault.

Offline overdraft

Pay-later credit that works with no network, inside the same signed vault as the customer's own money.

Cryptographic trust

Every payment is signed on the phone (ECDSA P-256) and hash-chained, so tampering and double-spends are detectable.

Any phone

Smartphone web app, USSD, SMS and printed stall stickers: no app store, no new hardware.

Offline sales as credit data

Trading days and takings, collected even without signal, become alternative data for lending decisions.

Delivery by any route

A payment reaches the bank through the trader's phone, the customer's phone, a nearby phone relaying it by QR, or SMS.

Pan-African by design

Country rules (currency, limits, ID types, rails) are configuration, not code. 15 markets set up, English and French.

WHY IT'S SAFE TO PAY OFFLINE

We don't claim double-spending is impossible. We make it bounded, detected and recovered.

Bounded

Small vault caps, per-payment and per-trader limits set per country. The worst case is a known, small amount.

Detected

Payments are numbered and hash-chained. A cloned phone shows up as a "fork" the moment payments sync.

Recovered

The trader is paid in full from a risk pool; the loss is recovered from the ID-verified holder. Lost phones can be frozen.

IMPACT

More sales, a credit history, and fairer loans.

Traders

Never lose a digital sale to bad network. Build a verifiable record. Access an overdraft to restock.

Customers

Pay digitally anywhere, on any phone, with small pay-later when short.

Partners

New, lower-risk SME lending backed by real sales data, plus transaction volume that used to be cash.

WHAT THE PILOT WILL MEASURE

- Sales captured offline
- Settlement rate
- Time to settle
- Fraud loss vs. volume
- Traders reaching credit
- Repayment rate
- Women-led share



BUSINESS MODEL

B2B infrastructure, white-labelled into our partners' apps.

Transaction fee

Small fee on each settled offline payment.

Credit share

Share of overdraft fees on credit PayVault data unlocks.

Platform licence

Per partner and per country, with integration support.

Risk pool

Optional guarantee premium covering offline fraud losses.

PayVault never holds customer funds. Licensed partners are the lender and custodian of record.

IMPLEMENTATION APPROACH

From working prototype to live pilot.

Done

Phase 0: Prototype

Live wallet, merchant mode, USSD, partner console, API and docs. Automated tests on payments, fraud and credit.

0–3 months

Phase 1: Pilot

One partner bank, MFB or mobile money operator and about 50 traders from one market association.

3–6 months

Phase 2: Harden

Android SDK, phone sign-in, independent security audit, hardware-secured keys, CBN regulatory sandbox.

6+ months

Phase 3: Scale

More partners and markets via country packs; cross-border settlement through PAPSS.

Integration points: partner core banking or wallet ledger (REST API + signed webhooks), KYC (NIN / BVN), SMS and USSD aggregator (Africa's Talking).

TECHNOLOGY

Built with proven, open technology.

App

Next.js / React Progressive Web App, TypeScript, service workers and IndexedDB for offline use.

Security

WebCrypto ECDSA P-256 signatures, non-extractable device keys, hash-chained payments, encrypted issuer keys.

Payments

Compact signed payment protocol over QR codes, PostgreSQL double-entry ledger and reconciliation.

Credit

Rules-based scoring on alternative data (trading days, sales, repayments); machine learning once pilot data exists.

Integration

REST APIs, HMAC-signed webhooks, Africa's Talking USSD and SMS.

Cloud

Docker, hosted in Lagos (RumptyCloud), with Vercel for the web front end.

TRY IT TODAY

Every trading day becomes credit history.

We're looking for a partner bank or mobile money operator and a market association to run the first pilot.

Live demo payvault.rumptycloud.app

Demo uses test money only. Country limits are illustrative until agreed with partners and regulators.

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